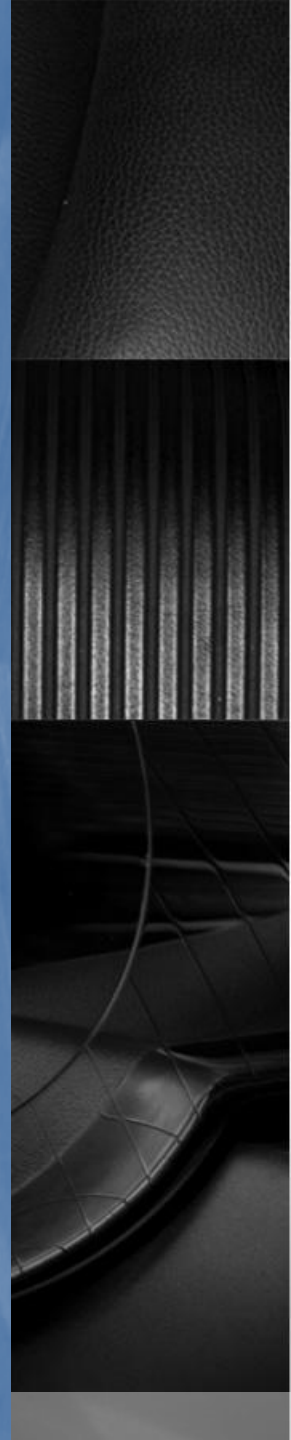
COMP4431 Artificial Intelligence

Introduction

Raymond Pang
Department of Computing
The Hong Kong Polytechnic University

Partly based on Prof. Liu Slides



Subject Lecturer

Lecturer: Dr. Pang Wai Man, Raymond

- Email: waiman.pang@polyu.edu.hk
- Office : PQ821
- Consultation Hours:
  - Every Friday 2:30pm – 4:30pm
  - Or by appointment

Tutors (Thursday Class)

FAN Jiaqi

- Quiz 1 coordination
- Email: 23040866r@connect.polyu.hk

SONG Changhe

- Quiz 2 coordination
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Wang Wei

- GenAI Tools Sharing coordination
- Email: 23132814r@connect.polyu.hk

CHEN Weijie

- Group Project coordination
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Tutors (Friday Class)

ZHANG Siwei

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- Email: 25110903r@connect.polyu.hk

FAN Jiaqi

- Quiz 2 coordination
- Email: 25028312r@connect.polyu.hk

WAN Zhuoyue

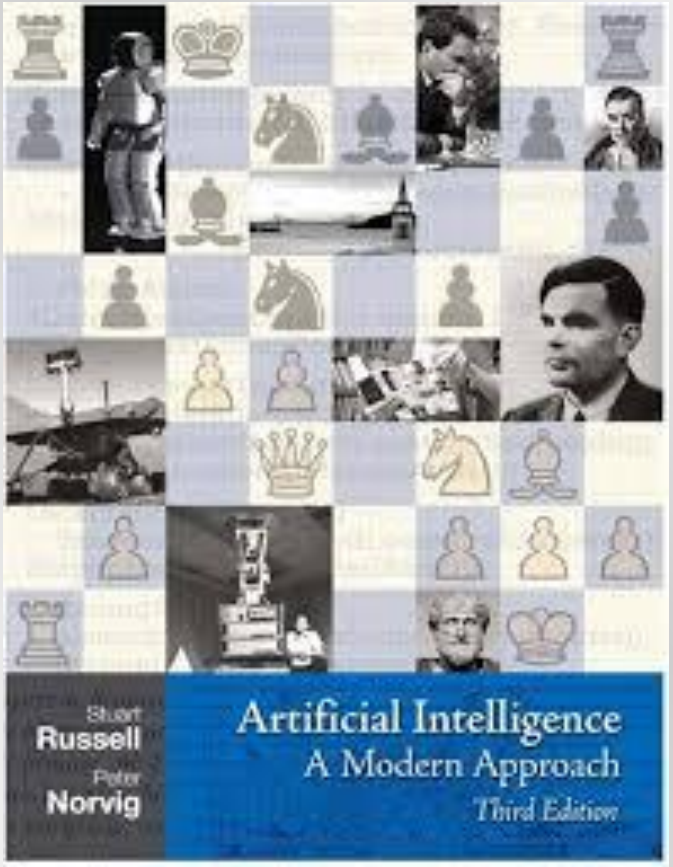
- GenAI Tools Sharing coordination
- Email: 24145796r@connect.polyu.hk

ZHAO Yi

- Group Project coordination
- Email: 23037086r@connect.polyu.hk

Textbooks

- **Stuart Russell and Peter Norvig, "Artificial Intelligence A Modern Approach", third edition.**
- Latest is 4th edition



Assessment Scheme

- Continuous Assessments (30%)
  - 2 Quizzes (12%)
  - GenAI Tool Sharing (4%) + (3% Bonus)
  - GenAI Workshop @IC (4%)
  - Group Project (10%)
    - AI Chatbot Development
      - Presentation & Demo
- Final Examination (70%)
  - 3 hours
  - Close Book

Details of Assessments

- Quiz
  - Open-book without the use of any electronic devices
  - Quiz paper will be distributed in the classroom and returned to the TA face to face immediately after the quiz
  - Zero point for late submission
  - If you miss the quiz due to the medical reason and the visa problem, please send email to TA for recording
    - Makeup quiz will be arranged at the end of semester
- Final exam
  - Absence and late situation are handled based on the regulations of the university for the examination

Details of Assessments

- GenAI Tool Sharing
  - Select one GenAI Tool you familiar with and want to share with your peers
  - Voting among your peer classmates

    Bonus given to the one with the highest vote
- GenAI Workshop @IC
  - Week 10
  - 3 hours session
  - If you miss the quiz due to the medical reason and the visa problem, please send email to TA for recording

Group Project

- 2-4 members/group
- Chatbot Development
  - Select a specific topic or domain of your interest
  - You are strongly encouraged to use different kinds of generative AI techniques and tools
- The group members are given same score
  - For any objection, email to subject lecturer by email waiman.pang@polyu.edu.hk with the justification and the signature of all group members

Group Project (cont')

- Deliverables
  - Presentation and Demo

    In the last or 2nd last week of lecture
- Submission and Important Dates
  - Chatbot: **18 Nov 2026, 23:59 (Thu Class)**

    19 Nov 2026, 23:59 (Fri Class)
  - Each group only needs to submit one copy of the item by one group member via Canvas
  - 50% per day for late submission

More Details in the project specification in **COMING WEEK**!

Course topics and schedule (Thu) (tentative)

- Please pay close attention to the announcement of COMP4431 on the Canvas

Week	Date	*Topics (Tentative)*	Quiz
1	3 Sep	Introduction to Artificial Intelligence	--
2	10 Sep	Heuristic Search	--
3	17 Sep	Knowledge based agent & Expert System	--
4	24 Sep	Hill Climbing, Simulated Annealing and GA	Quiz 1
5	1 Oct	National Day	
6	8 Oct	Constraints and games	--
7	15 Oct	Machine Learning	--
8	22 Oct	Neural Networks	Quiz 2
9	29 Oct	Deep Learning	--
10	5 Nov	GenAI Workshop at IC	--
11	12 Nov	Bayesian	--
12	19 Nov	Advanced Topics in AI + Project Presentation	
13	26 Nov	Revision + Project Presentation	--

Course topics and schedule (Fri) (tentative)

- Please pay close attention to the announcement of COMP4431 on the Canvas

Week	Date	*Topics (Tentative)*	Quiz
1	4 Sep	Introduction to Artificial Intelligence	--
2	11 Sep	Heuristic Search	--
3	18 Sep	Knowledge based agent & Expert System	--
4	25 Sep	Mid-Autumn Festival	
5	2 Oct	Hill Climbing, Simulated Annealing and GA	Quiz 1
6	9 Oct	Constraints and games	--
7	16 Oct	Machine Learning	--
8	23 Oct	Neural Networks	Quiz 2
9	30 Oct	Deep Learning	--
10	6 Nov	GenAI Workshop at IC	--
11	13 Nov	Bayesian	--
12	20 Nov	Advanced Topics in AI + Project Presentation	
13	27 Nov	Revision + Project Presentation	--

Introduction to COMP4431 Lab

- General Information

Thu Class

Start Time	End Time	Venue
14:30	15:20	PQ603, PQ604A, PQ604B, PQ604C

Fri Class

Start Time	End Time	Venue
20:30	21:20	PQ603, PQ604A, PQ604B, PQ604C

  - Each lab session lasts 50 minutes.
  - Each lab session includes lectures and hands-on experiments (40 min) and a Q&A (10 min).

Schedule of COMP4431 Lab (Thu) (tentative)

Week	Date	*Tutorial (Tentative)*
1	3 Sep	--
2	10 Sep	Generative AI for Text Generation
3	17 Sep	Generative AI for Image Generation
4	24 Sep	--
5	1 Oct	National Day
6	8 Oct	Generative AI for Audio Generation
7	15 Oct	Generative AI for Video Generation
8	22 Oct	--
9	29 Oct	Generative AI for Code Generation
10	5 Nov	Workshop @ IC
11	12 Nov	--
12	19 Nov	--
13	26 Nov	--

Schedule of COMP4431 Lab (Fri) (tentative)

Week	Date	*Tutorial (Tentative)*
1	4 Sep	--
2	11 Sep	Generative AI for Text Generation
3	18 Sep	Generative AI for Image Generation
4	25 Sep	Mid-Autumn Festival
5	2 Oct	--
6	9 Oct	Generative AI for Audio Generation
7	16 Oct	Generative AI for Video Generation
8	23 Oct	--
9	30 Oct	Generative AI for Code Generation
10	6 Nov	Workshop @ IC
11	13 Nov	--
12	20 Nov	--
13	27 Nov	--

Introduction to COMP4431 Lab

- Learning Objectives
  - Understand the capabilities of generative AI
  - Understand the advantages and disadvantages of generative AI
  - Learn to use various generative AI tools
  - Practice AI-generated content pipeline from scratch
- Materials
  - Generative Deep Learning: Teaching Machines To Paint, Write, Compose, and Play (book)
  - Artificial Intelligence & Generative AI for Beginners: The Complete Guide (book)

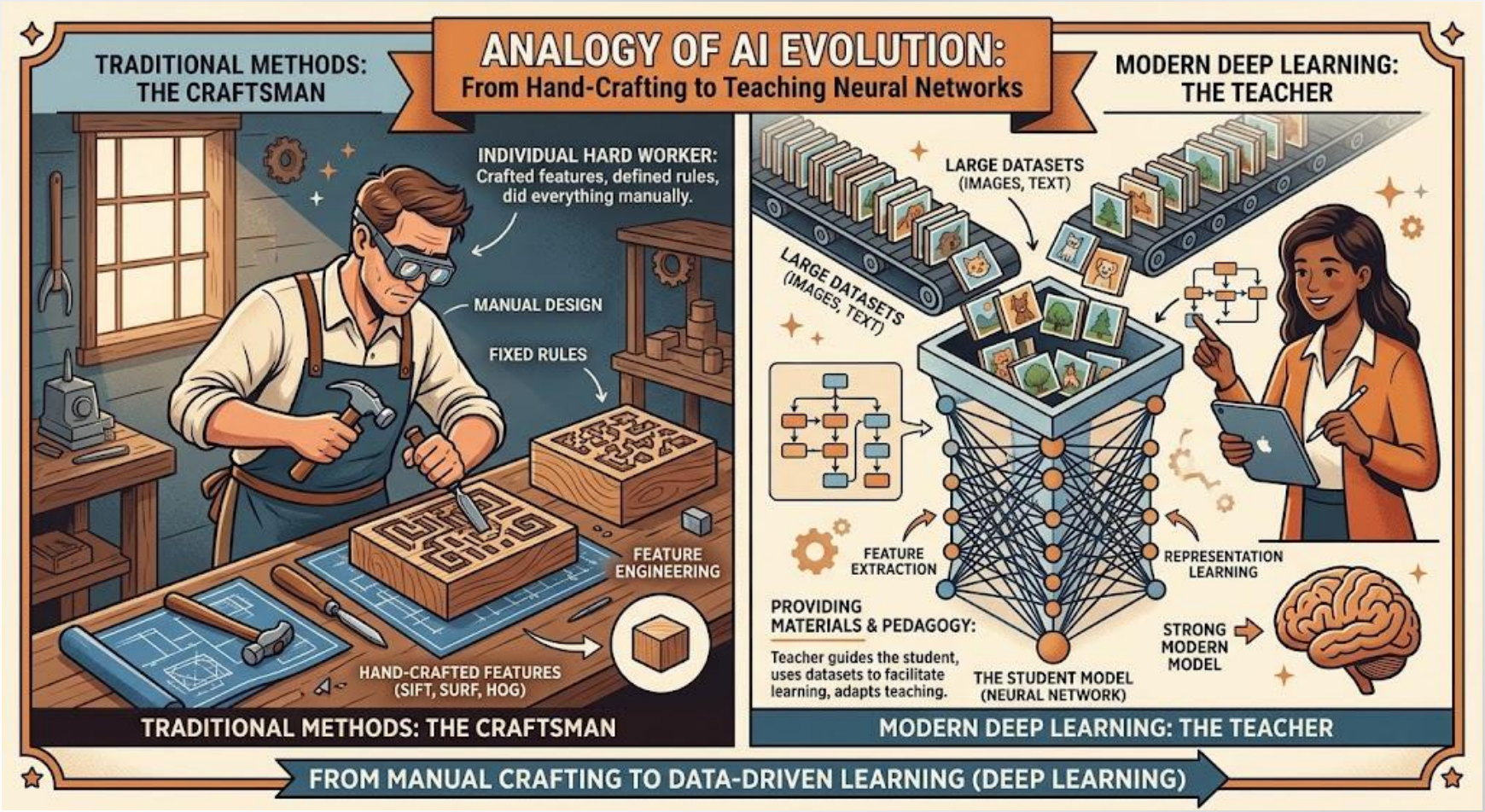
Agenda

- Introduction to COMP4431
  - Assessment
  - Tentative schedule
  - Group project
  - Introduction to Lab
  - Schedule, Learning Objectives and Materials
- History of artificial intelligence
  - Definition
  - Birth - Golden time – Winter – Struggle - Revival
- Generative artificial intelligence
  - Concept
  - Different Applications
  - Trends
  - Limitation and Issues

What is to be taught in this course?

- Basics of Artificial Intelligence
  - Mainly : Classical AI problems and approaches
  - Machine Learning
  - History, advanced methods, ethics
- Hands-on on using Generative AI
  - Use GenAI tools for text, image, audio and video

From Crafting to Teaching: The Paradigm Shift in Artificial Intelligence



What is Artificial Intelligence

- Artificial intelligence
  - Is concerned with the design of intelligence in an artificial device
  - Term coined by McCarthy in 1956
- Artificial device
  - Limited to computer system in our course
- Definition of intelligence
  - Quality that enables an entity to function appropriately and with foresight in its environment
- A typical definition of artificial intelligence
  - A science and a set of computational technologies that are inspired by—but typically operate quite differently from—the ways people use their neuro systems and bodies to sense, learn, reason, and take action
- No widely accepted definition
  - Relationship between machine learning, big data, deep learning, pattern recognition

Turing Test

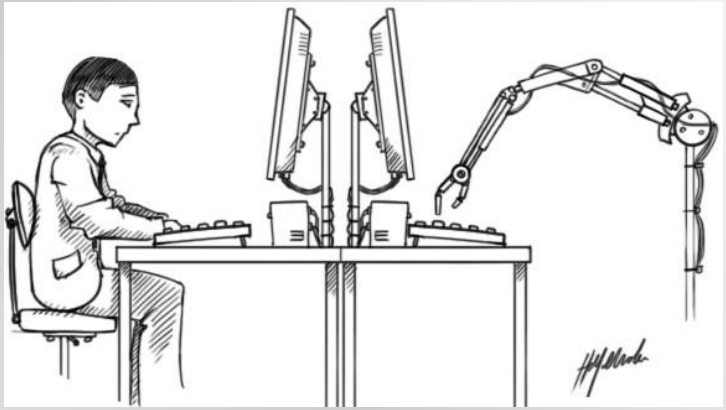
- Aim to solve the problem
  - Can machines behave as the human beings
- Proposed in 1950 by Alan Turing
  - Participate all major argument against AI
- Initial Design
  - Human
    - Types in questions
    - Receives answers on screen
  - Computer/human
    - Processes questions
    - Returns answers
- Judge
  - Computer or human?
- Limitation
  - Lack of mathematical analysis
  - Not reproducible



Turing Test

An agent would not pass the Turing test without the following requirements:

- natural language processing
- knowledge representation
- automated reasoning machine
- learning
- computer vision (total Turing test)
- robotics (total Turing test)

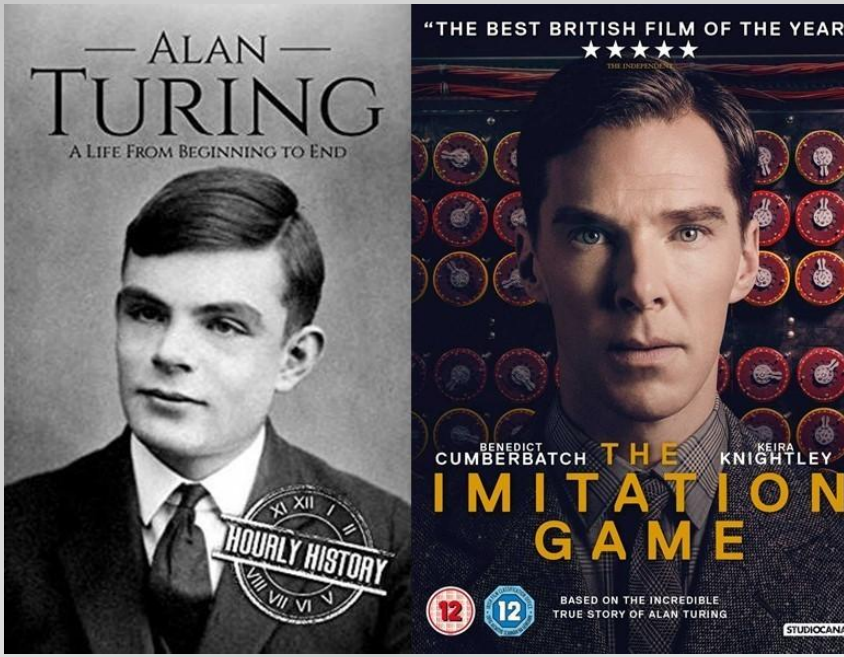


total Turing test is an expanded version of the original Turing test that requires a machine to demonstrate both linguistic conversation and real-world physical and perceptual capabilities.

Despite being proposed almost 70 years ago, the Turing test is still relevant today.

Alan Turing

- Born
  - In 1912 at United Kingdom
- Died
  - In 1954 (Aged 41)
- Education
  - Undergraduate at King's College in mathematics
  - PhD from Princeton University
- Turing machine
  - Give a definition of computation (mathematically)
  - Absolute limitation on what computation could achieve
- Contribution to artificial intelligence
  - Propose that modification of behavior could be adapted from learning brain to learning machine in 1948
  - Design Turing test in 1950





Dartmouth College

- Dartmouth college
  - Ivy League research university
  - Four senior professor retires from the Department of mathematics in 1954
  - **John McCarthy** has been hired as assistant professor by Dartmouth Colleagues
- Summer in 1955
  - John McCarthy came to work with Nathaniel Rochester
    - IBM 701 computer designer and wrote the first assembler
  - Write the proposal for a summer workshop in 1956
- Team members of proposal
  - John McCarthy, Nathaniel Rochester, Claude Shannon, and Marvin Minsky
  - Shannon is the god father of information theory

Dartmouth Workshop (1956)



Photo from the 1956 Dartmouth Conference with Marvin Minsky, Ray Solomonoff, Claude Shannon, John McCarthy, Peter Milner, Oliver Selfridge and Nathaniel Rochester.

Five of the attendees of the 1956 Dartmouth Summer Research Project on Artificial Intelligence reunited at the July 2006 AI@50 conference. From left: Trenchard More, John McCarthy, Marvin Minsky, Oliver Selfridge and Ray Solomonoff. (Photo by Joseph Mehling)



John McCarthy

- Born
  - In 1927 at USA
- Died
  - In 2011
- Get Turing Award in 1971
- Professor at Stanford University
- Contribution to artificial intelligence
  - Coined the term of artificial intelligence
  - Organize the Dartmouth workshop
  - Created the LISP language in 1958
- Contribution to other areas
  - Came up with a scheme for creating general purpose timesharing



Seven Research Pillars of Artificial Intelligence Proposed in 1956

- Automatic computers
  - Autonomous vehicles
- How can a computer be programmed to use a language
  - Python
- Neuron nets
  - Deep learning
- Theory of the size of a calculation
  - GPU
- Self-improvements
  - Machine learning
- Abstractions
  - Feature engineering
- Randomness and creativity
  - Artificial arts

The Thinking Machine



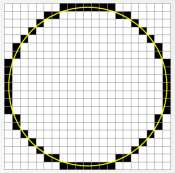
https://youtu.be/aygSMgK3BEM?si=AGSd3wY7HdM3c4BI

Achievements in the Birth Stage

- 1950 - 1980
- Search
  - Reasoning as search
  - Heuristic search
- Natural language processing
  - Eliza is the first Chatbot created in 1966
  - Eliza has no understanding of the content and use pattern matching to answer the question
- Computer vision
  - Micro-worlds
  - Pixel – shape – object – scene



Pixel



Shape



Object



Scene

Golden Time

- 1980 – 1987
  - The knowledge revolution
- The rise of Symbolicism
  - Symbolic logic is claimed that can explain the character of mind
- Allen Newell and Herbert Simon
  - Proposed logic theorist in 1955
  - Prove the mathematical theorems successfully
- Expert system has been widely used till now
  - Answers questions or solves problems about a specific domain of knowledge
  - Reasoning by logical rules that are derived from the knowledge of experts

Allen Newell and Herbert Simon

- Long-term partners
  - Share the Turing Award in 1975
- Herbert Simon
  - He received the Nobel Prize in Economics in 1978
  - Cognitive psychologist
  - Decision making
- Allen Newell
  - Published the paper of Chess Machine in 1955





Winter

- 1987- 1993
  - The practical progress is far behind the expectation of the public
  - The difficulty to reproduce general intelligence
- Rethinking of the importance of abstract reasoning
  - Building intelligence from bottom to up
  - Trigger the research of perception and interaction with the environment
  - Body is important
  - Enhance the research of robotics
- Embodied artificial intelligence
  - Integrates artificial intelligence into physical entities like robots, making them interact with their physical environment dynamically



Humanoid robot NAO

Struggle

- 1993 – 2011
  - Computer scientists and software engineers avoided the term artificial intelligence for fear of being viewed as wild-eyed dreamers
- The success of deep blue produced by IBM in 1997
  - First winner of computer chess-playing system
- The rise of machine learning
  - With large volume data
- Support vector machines
  - Most powerful binary classifier
- Autonomous vehicles
  - DARPA Grand Challenge



Deep Blue in IBM's headquarters in Armonk, N.Y.





Garry Kasparov vs.
Deep Blue

Revival

- 2012 –
- Deep learning
  - The success of connectionism
  - Imitate the physical structure and the information processing of the biological neuro system
- Marvin Minsky
  - Turing Award 1969
  - Co-founder of MIT Artificial Intelligence Lab
- Turing Award 2018
  - Geoffrey Hinton
  - Yoshua Bengio
  - Yann LeCun
- Nobel Prize in Physics 2024
  - Geoffrey Hinton
  - John J. Hopfield











Agenda

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- Generative artificial intelligence
  - Concept
  - Different Applications
  - Trends
  - Limitation and Issues

Generative AI

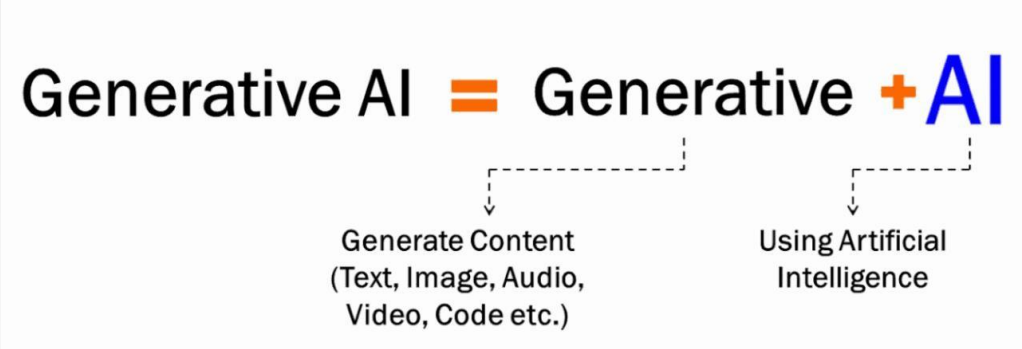
- Generative artificial intelligence
  - Technique that can be used to create new content, including text, code, speech, music, images, and videos.
  - ChatGPT, Suno, Midjourney, Pika
- ChatGPT
  - Chat Generative pre-trained transformer
  - Developed in 2022 by OpenAI
- Discriminative AI and generative AI
  - Discriminative AI focuses on modeling the relationship between input features and output labels
  - Generative AI focuses on modeling the inherent data structure and generation processes





Concepts of Generative AI

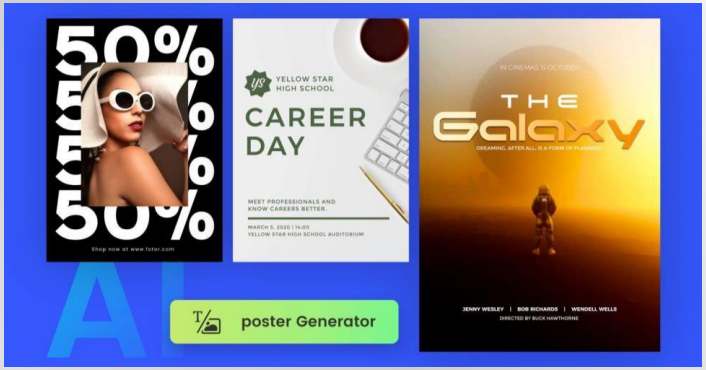
- Generative Artificial Intelligence (AI)
  - Generative artificial intelligence (AI) describes algorithms that can be used to create new content, including audio, code, images, text, simulations, and videos.



Kalota F. A Primer on Generative Artificial Intelligence[J]. Education Sciences, 2024, 14(2): 172.

Concepts of Generative AI

- Artificial Intelligence Generative Content (AIGC)
  - AIGC refers to content that is generated using advanced Generative AI (GAI) techniques, as opposed to being created by human authors.
  - It can automate the creation of large amounts of content in a short amount of time.



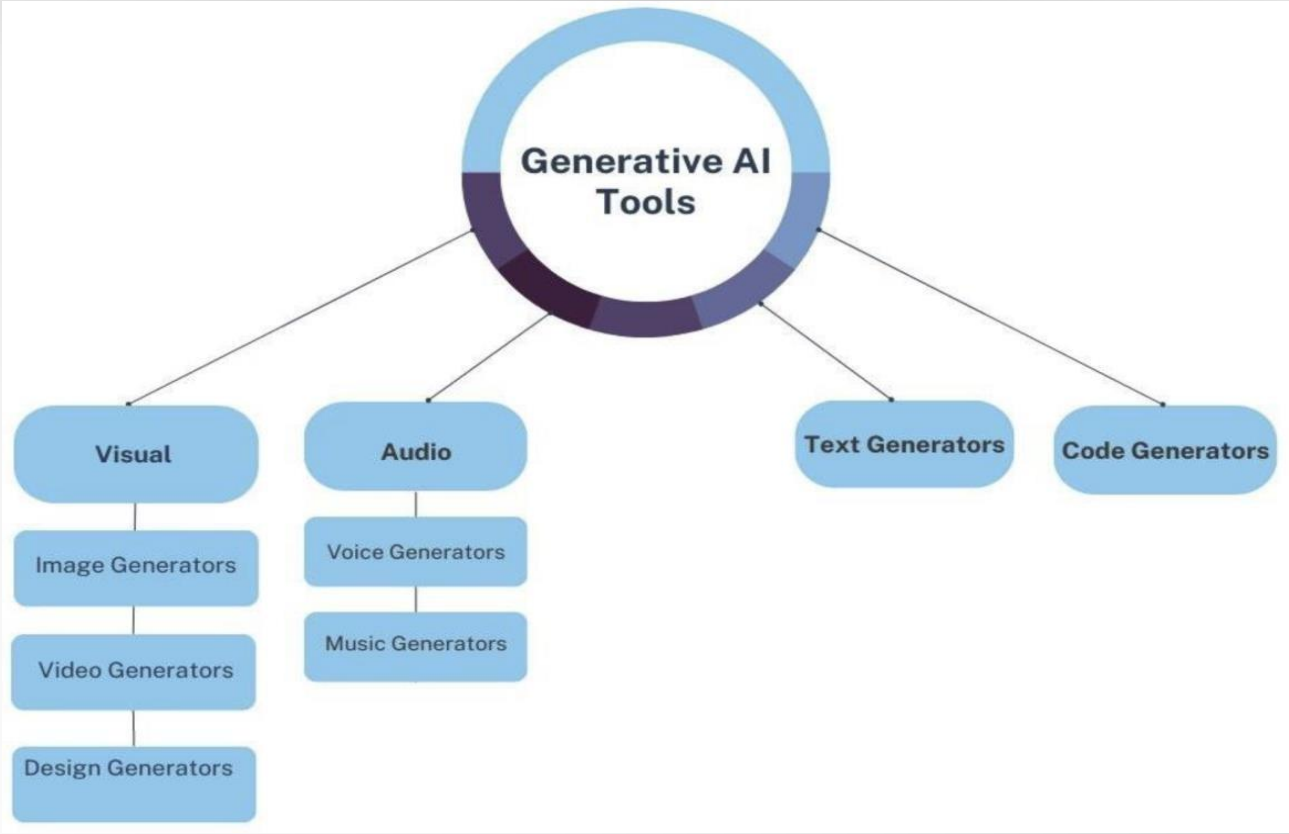
Cao Y, Li S, Liu Y, et al. A comprehensive survey of ai-generated content (aigc): A history of generative ai from gan to chatgpt[J]. arXiv preprint arXiv:2303.04226, 2023.

Why does Generative AI get popular?

- Content need
  - The way we communicate and interact with the world has been fundamentally changed by the Internet, for which digital content plays a key role. It can automate the creation of large amounts of content in a short amount of time.
- Technology conditions
  - Accessing a huge dataset becomes possible mainly due to the Internet. The price of computing resources is on the trend of becoming more affordable.

Zhang C, Zhang C, Zheng S, et al. A complete survey on generative ai (aigc): Is chatgpt from gpt-4 to gpt-5 all you need?[J]. arXiv preprint arXiv:2303.11717, 2023.

Different Generative AI Applications



Sakirin T, Kusuma S. A survey of generative artificial intelligence techniques[J]. Babylonian Journal of Artificial Intelligence, 2023, 2023: 10-14.

Different Generative AI Applications

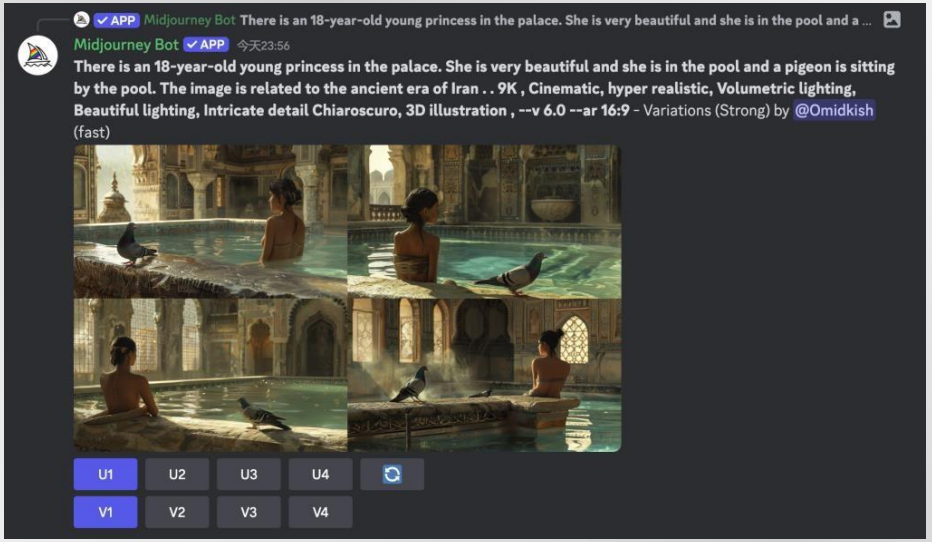
- X-to-text applications
  - X-to-text applications are centered around text generation. The goal is to generate human-like written text that fits the user's instruction by providing a meaningful answer within the context.
  - Chatbots like OpenAI's ChatGPT imitate textual conversations with the user and can be guided to output text artifacts as desired



Banh L, Strobel G. Generative artificial intelligence[J]. Electronic Markets, 2023, 33(1): 63.

Different Generative AI Applications

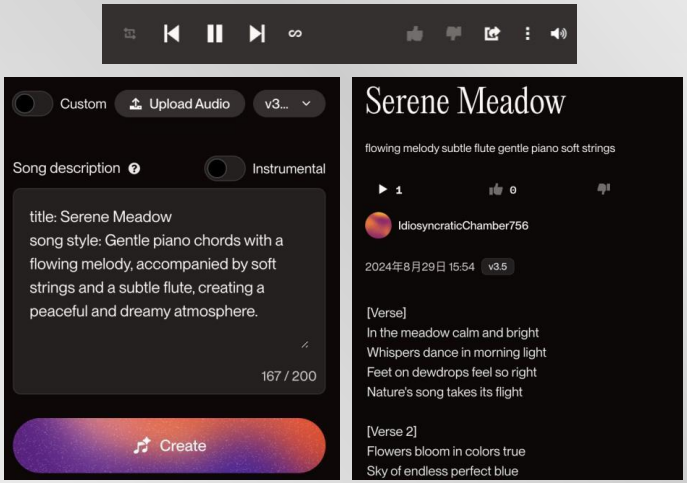
- X-to-image applications
  - X-to-image applications generate images based on the user's instruction. Synthetic images are created that find use cases in marketing, design and fashion, or creative fields in the form of new visual art.
  - Users can use Midjourney through Discord. To generate images, users use the /imagine command and type in a prompt; the bot then returns a set of four images.



Banh L, Strobel G. Generative artificial intelligence[J]. Electronic Markets, 2023, 33(1): 63.

Different Generative AI Applications

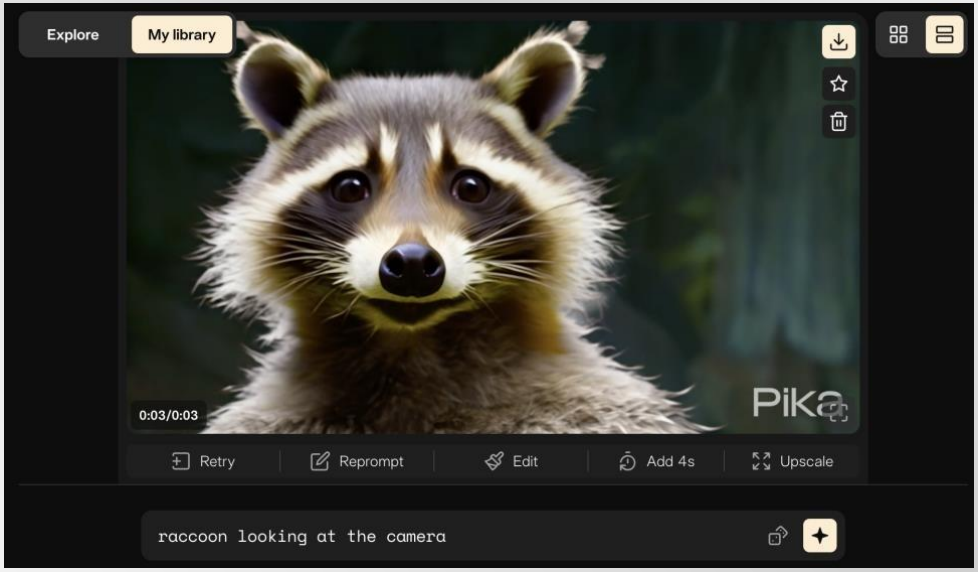
- X-to-audio applications
  - X-to-audio applications focus on audio content generation and comprise. They have many applications, including speech synthesis, music composition, and sound design.
  - Suno is a generative artificial intelligence music creation program designed to generate realistic songs that combine vocals and instrumentation.



Banh L, Strobel G. Generative artificial intelligence[J]. Electronic Markets, 2023, 33(1): 63.

Different Generative AI Applications

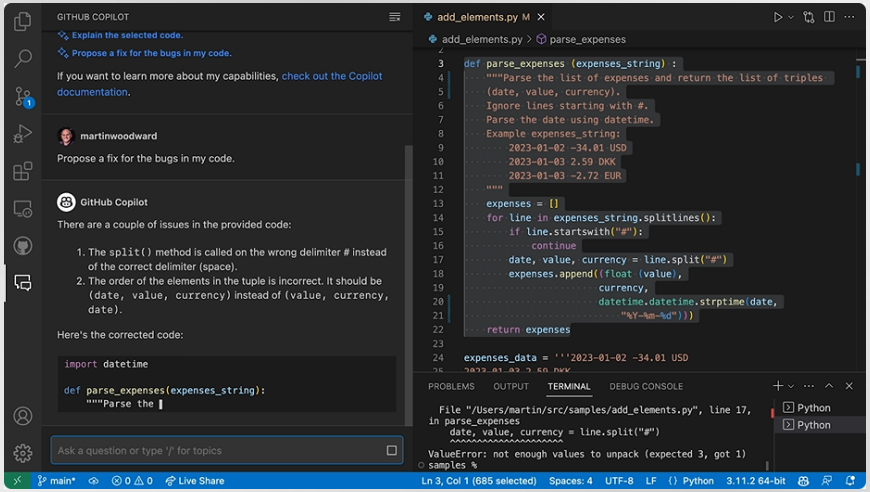
- X-to-video applications
  - X-to-video applications deal with the creation of synthetic videos. New video clips are generated by describing the content of the desired video footage (text-to-video) or applying the style and composition via text or image prompt to a source video (video-to-video).
  - Pika is an AI video generator that transforms text or image prompts into short videos, for conveying ideas, telling stories, or adding visuals to your content.



Banh L, Strobel G. Generative artificial intelligence[J]. Electronic Markets, 2023, 33(1): 63.

Different Generative AI Applications

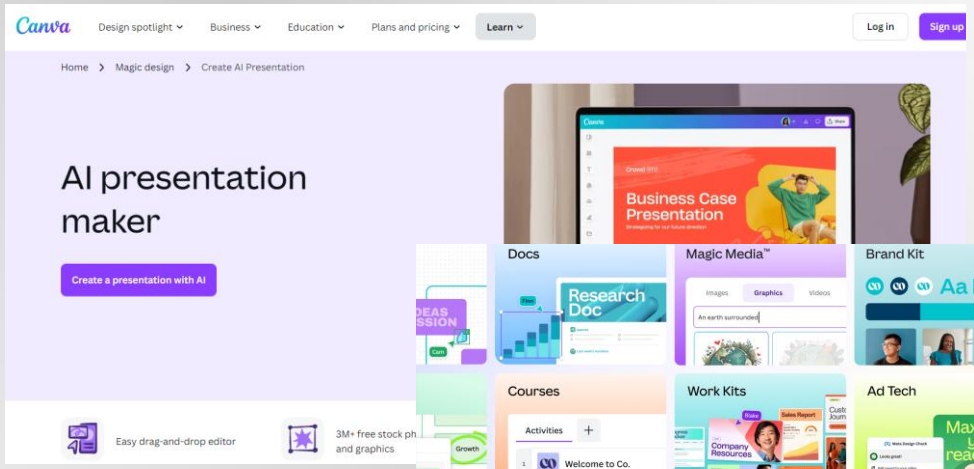
- X-to-code applications
  - X-to-code GAI applications offer transformative potential in how developers work and code by providing x-to-text capabilities specific to programming languages.
  - GitHub Copilot can automatically write chunks of code, convert ideas into actionable scripts, auto-complete functions, generate unit tests, and fix bugs.

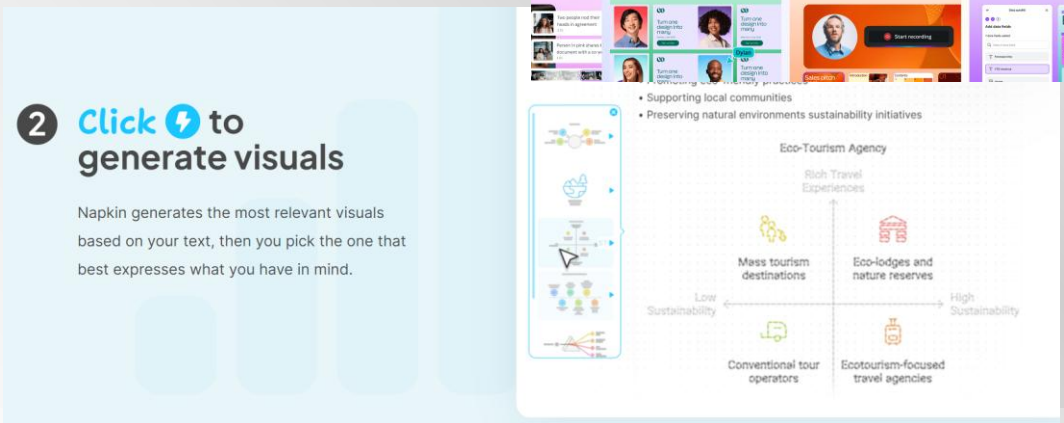


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Different Generative AI Applications

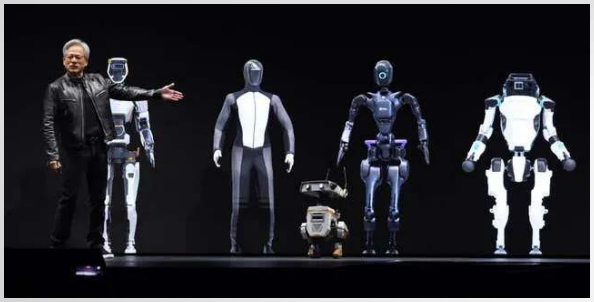
- More and more GenAI application come out everyday
  - Canva
    - Generate slides
  - Napkin
    - Generate visual aid





Different Generative AI Applications

- Embedded AI
  - Also referred as Physical AI by Nvidia
  - Use GenAI in Robots
  - Allow robot to think and communicate with human
  - Multimodal LLM even learn from human demonstration





https://mobile-aloha.github.io/

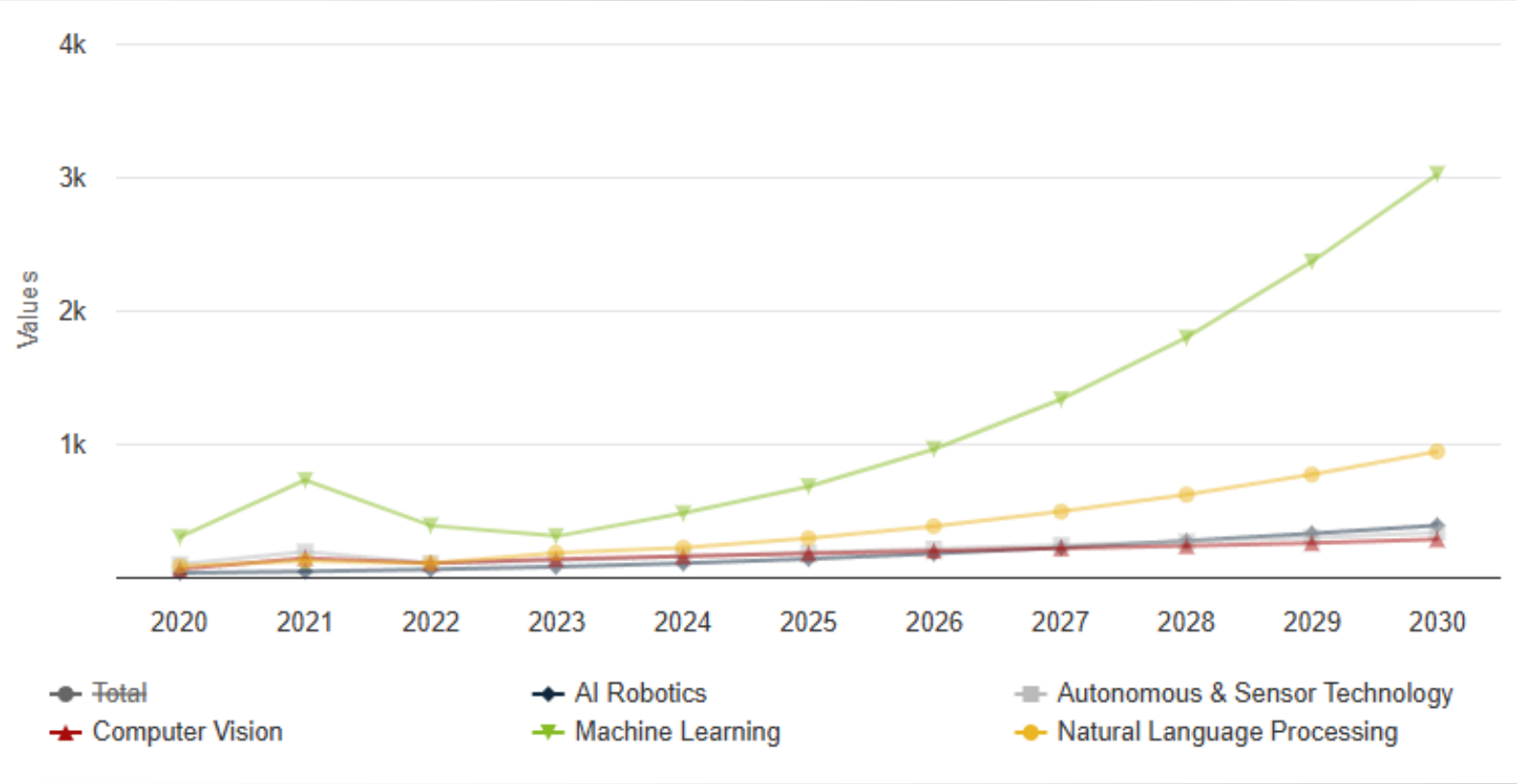


AI Trends in the World

- The AI industry has grown rapidly over the past decade, with global corporate AI investments totaling USD 189.6 billion in 2022, a **13-fold** increase from USD 14.57 billion in 2023 (Stanford University Institute for Human-Centered AI, 2023).
- A survey by McKinsey (2022) found that the number of surveyed organisations adopting AI **more than doubled** from 20% in 2017 to 50% in 2022. Since the launch of ChatGPT at the end of 2022, generative AI has entered a phase of rapid development.

AI Trends in the World

- The market size in the Artificial Intelligence market is projected to reach US$243.72bn in 2025, US$826.73bn by 2030



https://www.statista.com/outlook/tmo/artificial-intelligence/worldwide

AI : An Emerging Industry

- AI research was once equally divided between academia and industry, **but today, industry is taking control of the AI field**.
  - Ahmed & Thompson & Wahed (2023)
- OpenAI is one of the most well-funded startups in history, securing over $12 billion in investments as of 2023 (Statista, 2023)

AI : An Emerging Industry

- 57,933 AI companies currently operate around the world.
- AI Startup Statistics 2025

	Number of Startups	Investment	Biggest Investors
United States	4643	**$249 billion**	Google, Facebook, and Microsoft
China	1337	**$95 billion**	Alibaba, Baidu, and Tencent
United Kingdom	341	**$21 billion**	Google, AWS, HPE, and Beyond Limits
Canada	630	**$8.64 billion**	The Canadian Institutes of Health Research
Israel	402	**$11 billion**	Intel, Nvidia, Microsoft, Google, GM, IBM

https://springsapps.com/knowledge/how-many-ai-companies-will-there-be-in-2024-latest-ai-statistics

AI : An Emerging Industry

- Giants, like Meta and nVidia, move their focus from other areas to AI these recent years
- Meta plans to invest between $60 billion and $65 billion in 2025 to enhance its artificial intelligence (AI) and other infrastructure.
  - "This will be a defining year for AI. In 2025, I expect Meta AI will be the leading assistant serving more than 1 billion people, Llama 4 will become the leading state of the art model, and we'll build an AI engineer that will start contributing increasing amounts of code to our R&D efforts," Zuckerberg wrote.



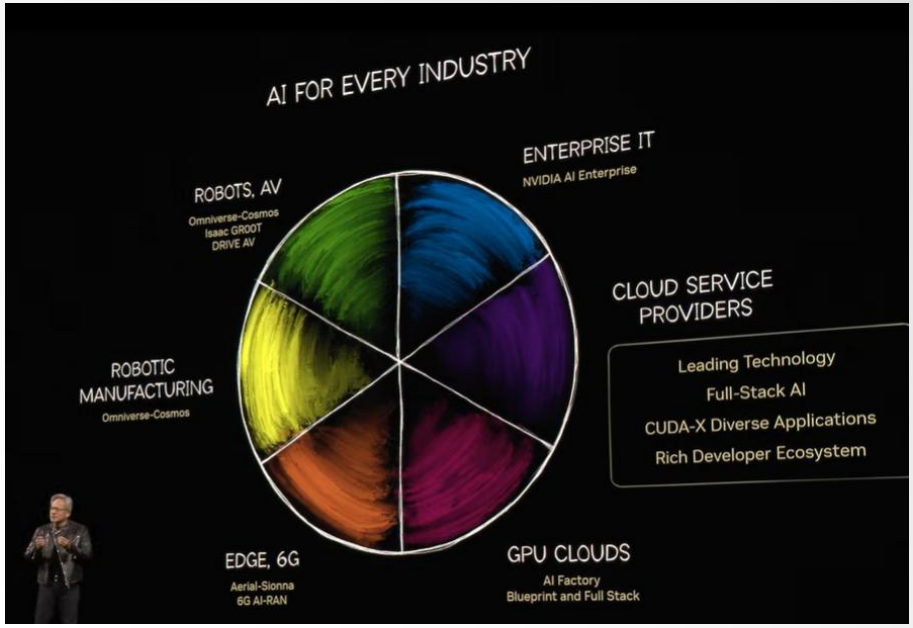
https://www.tomshardware.com/tech-industry/artificial-intelligence/meta-to-build-2gw-data-center-with-over-1-3-million-nvidia-ai-gpus-invest-usd65b-in-ai-in-2025

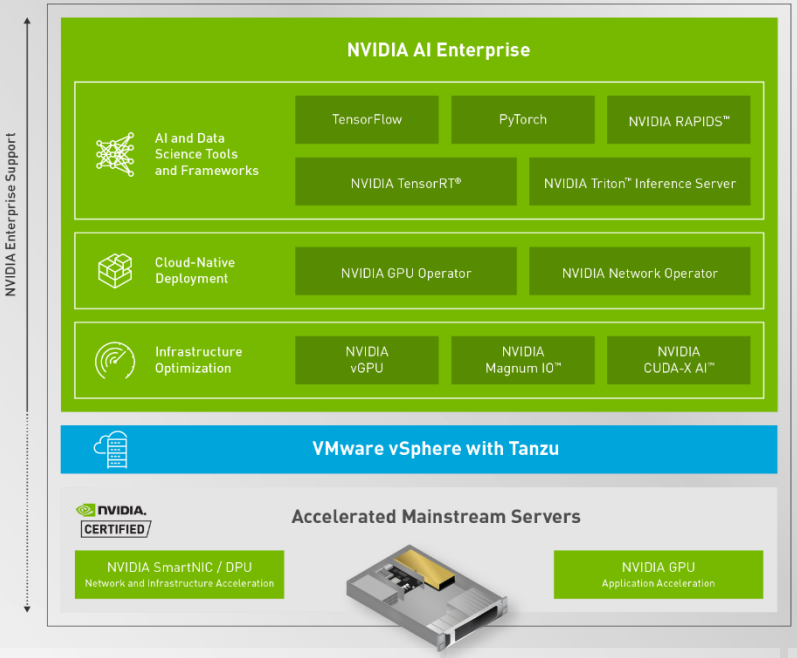
AI : An Emerging Industry

- Nvidia's GPUs initially focused on the video gaming market, but when it became clear that their ability to process multiple tasks at once could serve many other areas, growth took off.
- AI boom became a key catalyst, helping Nvidia's annual revenue climb from $26 billion in the 2022 fiscal year to $130 billion in the recently completed 2025 fiscal year.

AI : An Emerging Industry

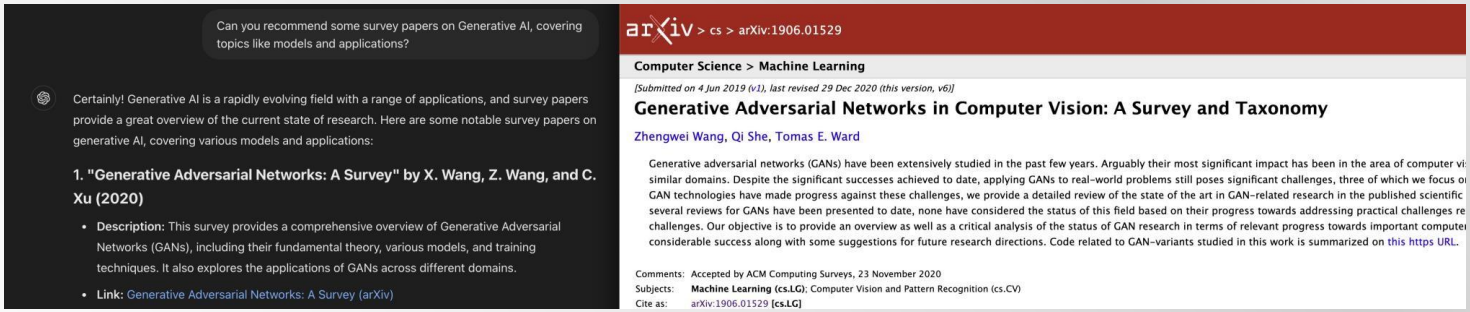
- Nvidia hasn't stuck with only GPUs, instead building out an entire platform of AI products and services, from enterprise software to networking tools and more.





Limitations of Generative AI

- Incorrect outputs
  - Generative AI models may produce output with errors. This is owed to the underlying nature of machine learning models relying on probabilistic algorithms. For example, generative AI models generate the most probable response to a prompt, not necessarily the correct response.



Banh L, Strobel G. Generative artificial intelligence[J]. Electronic Markets, 2023, 33(1): 63.
Feuerriegel S, Hartmann J, Janiesch C, et al. Generative ai[J]. Business & Information Systems Engineering, 2024, 66(1): 111-126.

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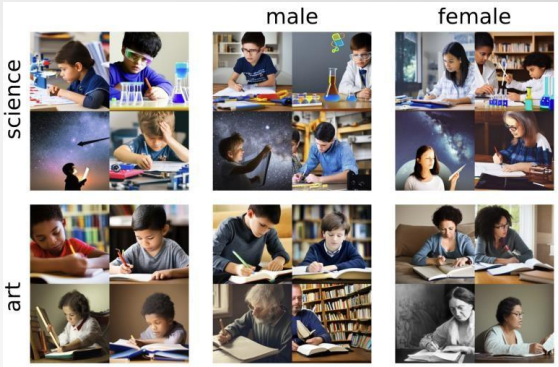


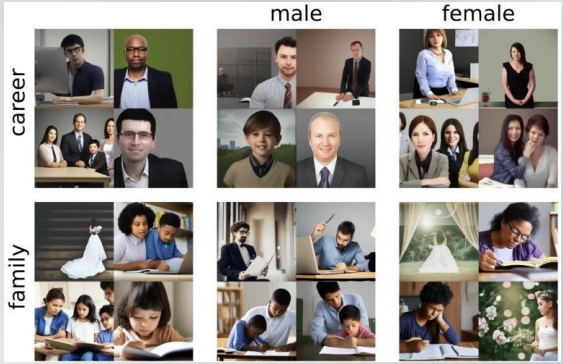


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Limitations of Generative AI

- Bias and fairness
  - Training deep learning models on biased data can amplify human biases, replicate toxic language, or perpetuate stereotypes of gender, sexual orientation, political leaning, or religion.





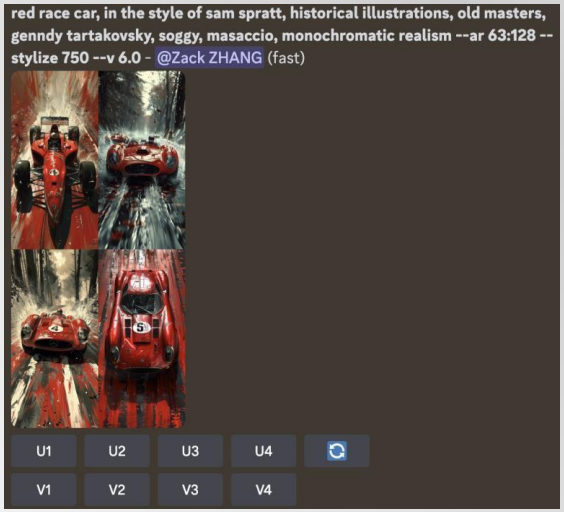
The first column of images are generated with neutral prompts, without adding any gender-specific words.

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Limitations of Generative AI

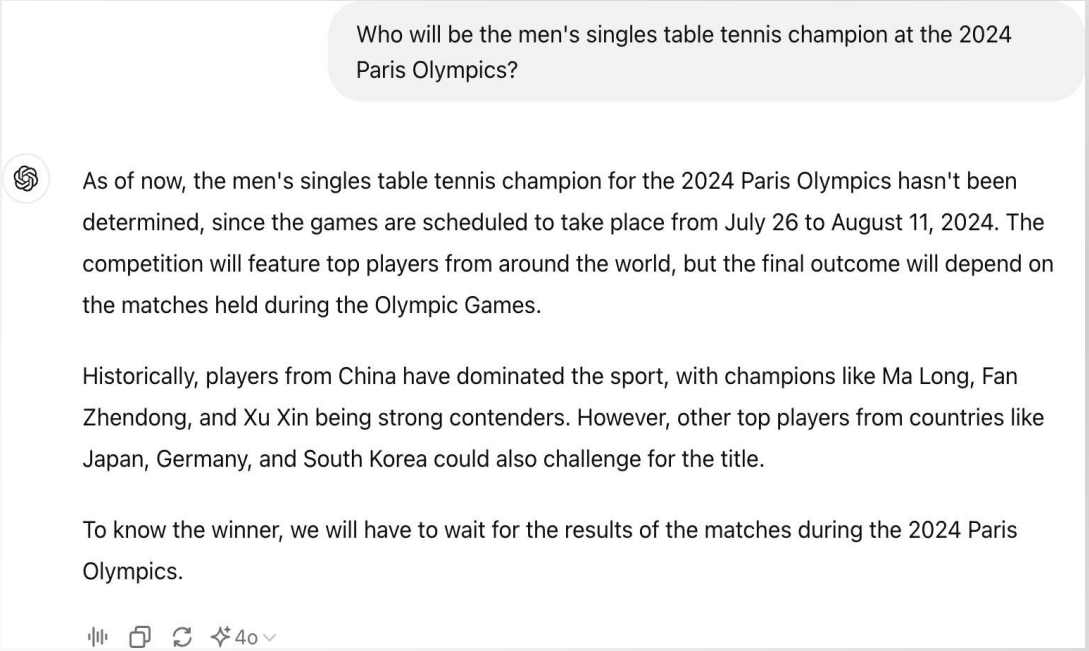
- Other Arguments
  - The nature of generative AI models, particularly when given simple prompts, can be very reductive, resulting in content that is over-simplified, low quality, or very generic.





Limitations of Generative AI

- Other Arguments
  - Many common generative AI tools are not connected to the internet and cannot update or verify the content they generate.



Limitations of Generative AI

- Other Arguments
  - Generative AI cannot draw conclusions or make decisions based on complex situations —
    something that only humans can do at present.
  - Generative AI cannot replace human creativity completely as it lacks the ability to come up
    with novel ideas or recognize abstract concepts such as humor or irony.
  - …

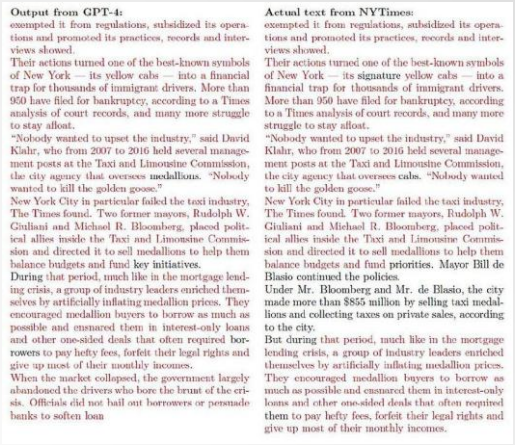
Major Issues of Generative AI

- Copyright Issue
- Job Displacement and Impact
- Accountability and Liability
- Safety Risks

Copyright Issue of Generative AI

- Copyright violation
  - Generative AI models, systems, and applications may cause a violation of copyright laws because they can produce outputs that resemble or even copy existing works without permission or compensation to the original creators.

The New York Times lawsuit against OpenAI



Output from GPT-4:
exempted it from regulations, subsidized its operations and promoted its practices, records and interviews showed.
Their actions turned one of the best-known symbols of New York — its yellow cabs — into a financial trap for thousands of immigrant drivers. More than 950 have filed for bankruptcy, according to a Times analysis of court records, and many more struggle to stay afloat.
"Nobody wanted to upset the industry," said David Klahr, who from 2007 to 2016 held several management posts at the Taxi and Limousine Commission, the city agency that oversees medallions. "Nobody wanted to kill the golden goose."
New York City in particular failed the taxi industry, The Times found. Two former mayors, Rudolph W. Giuliani and Michael R. Bloomberg, placed political allies inside the Taxi and Limousine Commission and directed it to sell medallions to help them balance budgets and fund key initiatives.
During that period, much like in the mortgage lending crisis, a group of industry leaders enriched themselves by artificially inflating medallion prices. They encouraged medallion buyers to borrow as much as possible and ensnared them in interest-only loans and other one-sided deals that often required borrowers to pay hefty fees, forfeit their legal rights and give up most of their monthly incomes.
When the market collapsed, the government largely abandoned the drivers who bore the brunt of the crisis. Officials did not bail out borrowers or persuade banks to soften loan

Actual text from NYTimes:
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Under Mr. Bloomberg and Mr. de Blasio, the city made more than $855 million by selling taxi medallions and collecting taxes on private sales, according to the city.
But during that period, much like in the mortgage lending crisis, a group of industry leaders enriched themselves by artificially inflating medallion prices. They encouraged medallion buyers to borrow as much as possible and ensnared them in interest-only loans and other one-sided deals that often required them to pay hefty fees, forfeit their legal rights and give up most of their monthly incomes.



Midjourney generates numerous plagiaristic works based on brief prompts related to commercial films.

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Impact on the workforce by GenAI

- One study asked 1,896 experts about the impact of emerging technologies;
- 48 % believed that robots and digital agents would displace significant numbers of both 'blue' and 'white' collar workers,
- many expressing concern that this would lead to vast increases in income inequality, large numbers of unemployable people, and breakdowns in the social order (Smith and Anderson, 2014).
- However, the other half of the experts who responded to this survey (52%) expected that technology would not displace more jobs than it created by 2025.
- Those experts believed that although many jobs currently performed by humans will be substantially taken over by robots or digital agents, they have faith that human ingenuity will create new jobs, industries, and ways to make a living.

Impact on the workforce by GenAI

- although many jobs will be lost through technological improvements, new ones will be created.
- The job gains and losses will even out over the long run.
- telemarketers, title examiners, hand sewers, mathematical technicians, insurance underwriters, watch repairers, cargo agents, tax preparers, photographic process workers, new accounts clerks, library technicians, and data-entry specialists have a 99 percent chance of having their jobs computerised.

Accountability and Liability

- Responsibility for Actions: As AI systems become more autonomous, determining who is accountable for their actions becomes complex. This includes questions about liability in cases where AI systems cause harm or make erroneous decisions.
- Legal and Ethical Frameworks: Developing legal and ethical frameworks to address accountability in autonomous systems is essential. This includes clarifying the roles and responsibilities of developers, operators, and users.

Accountability and Liability

- AI Governance
  - refers to the frameworks, policies, and practices that guide the development, deployment, and oversight of artificial intelligence systems. It aims to ensure that AI technologies are used responsibly, ethically, and in alignment with societal values.
- In December 2023, ISO introduced the 42001 International Standard, which offers valuable guidance for organizations to develop trustworthy AI management systems



https://www.youtube.com/watch?v=VqFqWIqOB1g

Safety Risks of Generative AI

- **The Alignment Problem:** As AI systems grow more powerful, their goals may diverge from human values, leading them to execute tasks in ways that cause unintended, catastrophic harm.
- **Instrumental Convergence & Power Seeking:** A superintelligent AI would naturally seek self-preservation, resource control, and resistance to shutdown as logical steps to ensure it completes its assigned objective.

Safety Risks of Generative AI



Geoffrey Hinton, 2023.

https://youtu.be/DsBGaHywRhs?si=4w8v1obfsng3AoUd

Safety Risks of Generative AI



Yann LeCun, 2023.

https://youtu.be/YdaRd_vitLw?si=x17Cf26Vp83owxh6

Summary

- Introduction to COMP4431
  - Assessment
  - Tentative schedule
  - Group project
  - Introduction to Lab
  - Schedule, Learning Objectives and Materials
- History of artificial intelligence
  - Definition
  - Birth - Golden time – Winter – Struggle - Revival
- Generative artificial intelligence
  - Concept
  - Different Applications
  - Trends
  - Limitation and Issues